BST INTERNAL HACKTHAON 2026

Basic Details

- Problem Statement ID -
- Problem Statement Title- **Street - level civic and road hazard tracker**

- Theme- **Smart Cities and Urban Development**

- PS Category- **Software**

- Team ID -

- Team Name – **Glitch Hunters**

GLITCH HUNTER STREET-LEVEL CIVIC HAZARD TRACKER

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Challenges and Problems

- **Potholes & Road Damage**
  Damaged roads can cause accidents, vehicle damage and traffic disruption.
- **Broken Streetlights**
  Dark or poorly lit streets reduce safety for pedestrians and commuters.
- **Open Drains & Manholes**
  Uncovered civic infrastructure creates serious street-level hazards.
- **Waste & Illegal Dumping**
  Overflowing garbage affects cleanliness, health and neighborhood safety.
- **Waterlogging & Blocked Footpaths**
  Poor drainage and damaged walkways make streets difficult to use.

Innovation and Uniqueness

AI Hazard Detection
Analyze uploaded images to help identify hazard types.

GPS Mapping
Pin the exact hazard location and display it on a live map.

Smart Prioritization
Rank hazards by severity, location and public impact.

Duplicate Detection
Detect similar reports to reduce repeated complaints.

Citizen Engagement
Simple reporting, status updates and community visibility.

Authority Dashboard
Track, assign and update civic issues from one place.

Proposed Solution

Glitch Hunter is a digital platform that helps citizens report, locate and track street-level civic hazards.

- Report a hazard with photo, category, severity and description.
- Automatically capture the location using GPS.
- View reported hazards on a map and check their status.
- Help authorities prioritize and assign issues for resolution.
- Receive updates from report submission to resolution.

How it Addresses the Problems

Faster Reporting	One place to submit hazards in seconds.
Accurate Location	GPS reduces confusion about where the issue is.
Priority-Based Action	High-risk hazards can be highlighted first.
Transparent Tracking	Citizens can see reported, assigned and resolved status.
Safer Communities	Better visibility supports quicker civic action.

GLITCH HUNTER

See It • Report It • Fix It

TECHNICAL APPROACH

SMARTER REPORTING | SAFER STREETS | STRONGER COMMUNITIES

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Methodology and Process

1 Data Collection & Integration
- Gather real-time data from user reports, photos, GPS location and integrate with open government datasets (e.g., maps, civic data, local authorities).

2 Data Preprocessing & Storage
- Clean and standardize data, remove duplicates, and store securely in a scalable cloud database.

3 AI/ML Model Development
- Use ML models to classify hazards, prioritize by severity, and predict risk areas based on patterns.

4 Blockchain & Verification
- Verify reports, ensure data authenticity, and enable secure, transparent tracking of complaints.

5 Immersive & Real-Time Services
- Provide AR/VR map views, live location tracking and real-time updates for users.

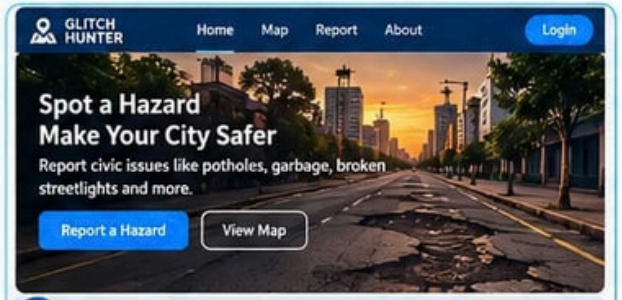
6 Scaling & Improvement
- Expand coverage, integrate more communities, and use user feedback to improve accuracy and services.

Technology Stack

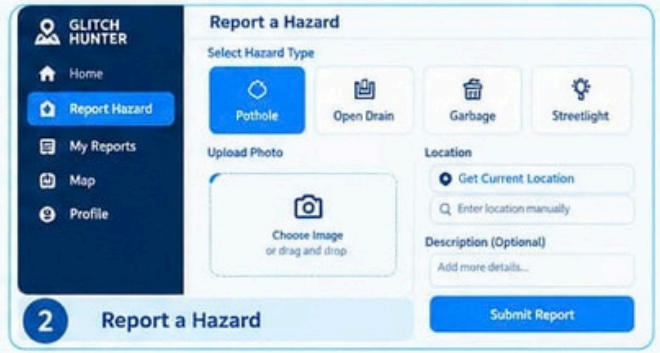
Python	FastAPI	JavaScript	HTML5	CSS3	Firebase
(Backend)	(Framework)	(Interactivity)	(Structure)	(Design)	(Realtime)

User Interface

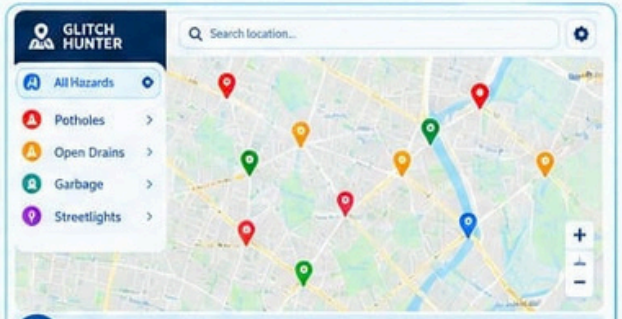
Glitch Hunter – Street-Level Civic Hazard Tracker



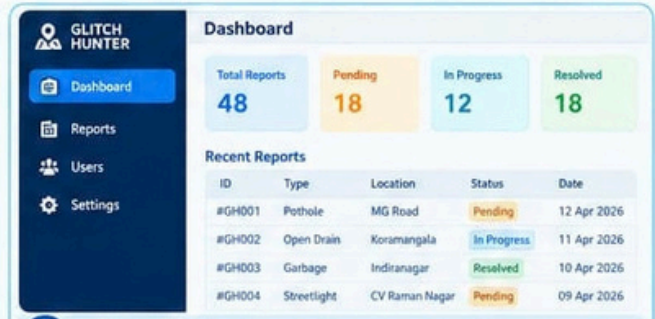
1 Home Page



2 Report a Hazard



3 Live Hazard Map



4 Admin Dashboard

Reference Apps & Websites

Swachhata App / Swachh City
- Civic complaints and status updates.
- https://dashboard.swachh.city/

Smart Cities Mission
- Urban infrastructure and citizen-focused solutions.
- https://www.mohua.gov.in/

CPGRAMS
- Centralized Public Grievance Redress and Monitoring System.
- https://pgportal.gov.in/

Research Articles & Sources

Smart Cities Mission
Ministry of Housing and Urban Affairs
https://www.mohua.gov.in/

Swachhata App – Citizen Complaint
Swachh Bharat Mission (Urban)
https://dashboard.swachh.city/

CPGRAMS
Government of India
https://pgportal.gov.in/

WHAT MAKES GLITCH HUNTER DIFFERENT?

- Location-based hazard reporting
- Photo evidence for civic issues
- Interactive street-level hazard map
- Complaint tracking and resolution updates
- Hazard prioritization for faster action

Street-Level
Civic Hazard Tracker
Report • Track • Safer Streets

FEASIBILITY AND VIABILITY

SMARTER REPORTING | SAFER COMMUNITIES

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Feasibility and Viability

Technical Feasibility

- Use of GPS, maps and mobile technology.
- Real-time data collection via user reports & images.
- Scalable and secure cloud infrastructure.

Financial Feasibility

- Government grants and smart city funds.
- Low cost mobile app & web platform development.
- Potential for CSR and public-private partnerships.

Operational Feasibility

- Integration with municipal and local authorities.
- Dedicated monitoring team for verification.
- Clear workflow for issue resolution and updates.

Social Viability

- Increases citizen participation and awareness.
- Builds trust between citizens and authorities.
- Empowers communities to report and solve local issues.

Blockchain & Security

- Secure and tamper-proof data records.
- Verified user reports and real-time updates.
- Protects user identity and prevents false reports.

Revenue Streams

- Government contracts and civic tech funding.
- Premium features for analytics and reports.
- Partnerships with urban development and safety programs.

Supply Chain & Logistics

- Direct linkage with municipal teams and maintenance units.
- Efficient allocation of resources for quick response.
- Tracking system for issue resolution progress.

Awareness & Adoption

- Promotes through social media, local campaigns and schools.
- Easy-to-use app and simple reporting process.
- Incentives for active citizens and volunteers.

Strategies for Overcoming Challenges

Technical Challenges

- **Issue:** GPS inaccuracy, poor internet connectivity and device limitations.
- **Solution:** Offline mode, map caching, optimized app design and multi-source data validation.

Financial Challenges

- **Issue:** Limited budget for maintenance and expansion.
- **Solution:** Government schemes (Smart Cities, AMRUT), CSR support and public-private partnerships.

Operational Challenges

- **Issue:** Delayed response, coordination issues and data verification.
- **Solution:** Dedicated monitoring team, clear SOPs, and integration with municipal systems.

Safer Roads | Cleaner Cities | Stronger Communities

GLITCH HUNTERS
HUNTING GLITCHES, BUILDING SMARTER CITIES!

Street-Level Civic Hazard Tracker

Report • Track • Safer Streets

IMPACT AND BENEFITS

SAFER STREETS | STRONGER COMMUNITIES | A SMARTER TOMORROW

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For Citizens
(General Public)

- Report hazards easily with photos and location.
- Get real-time updates and track resolution status.
- Feel safer and more informed in their daily commute.

For Government & Municipal Authorities

- Get real-time insights on civic issues.
- Prioritize and allocate resources efficiently.
- Improve planning and decision-making with data-driven reports.

BENEFITS OF THE SOLUTION

For Local Workers & Service Providers

- Receive timely alerts for quick action and maintenance.
- Improve response time and service efficiency.
- Coordinate better across departments.

For the Environment & Public Health

- Reduce risks from open drains, garbage and waterlogging.
- Prevent accidents and disease outbreaks.
- Support cleaner, safer and healthier communities.

For the Community & Society

- Build a culture of civic responsibility and participation.
- Encourage collective problem solving.
- Create safer, cleaner and more livable urban spaces.

For Urban Development & Future Planning

- Identify recurring issues and support long-term infrastructure planning.
- Enable data-driven smart city initiatives.
- Help build resilient and sustainable cities.

Target Audience

Citizens
(Students, residents, commuters)

Government
(Municipal bodies, ward offices)

Workers & Service Providers
(Municipal staff, utility teams)

Community
(Resident groups, NGOs, volunteers)

Value Added Services

Real-time Alerts
Instant notifications for new hazards and updates.

Interactive Map
View, report and track issues on a live map.

Data Analytics
Helps authorities identify patterns and plan better.

Mobile Friendly
Easy to use app and web platform.

Community Reporting
Verified and authentic reports from locals.

Multi-Department Integration
Connects with civic and emergency services.

Scalability

Expand to More Cities
From local areas to national level.

Increase User Base
Onboard more citizens, volunteers and stakeholders.

Add More Features
Traffic, waste, lighting, public utilities, etc.

Better Data → Faster Action → Safer Streets → Stronger Communities

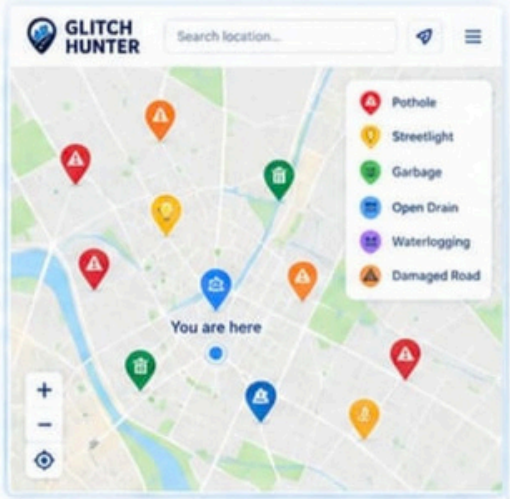
GLITCH HUNTER
Street-Level Civic Hazard Tracker

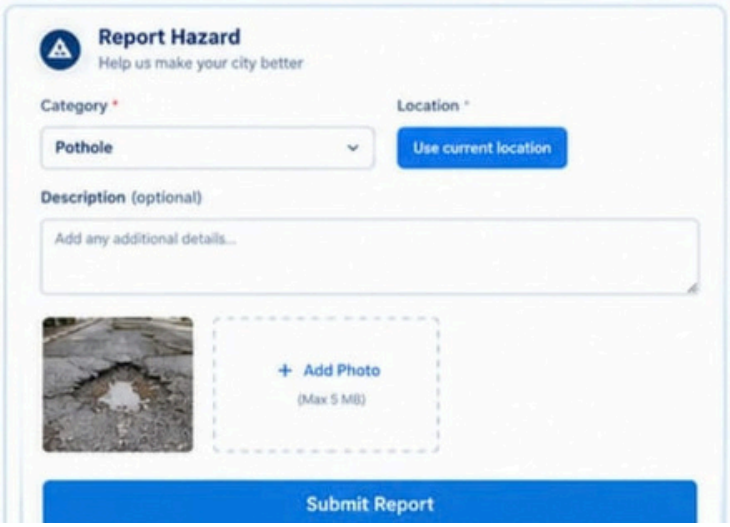
RESEARCH AND REFERENCES

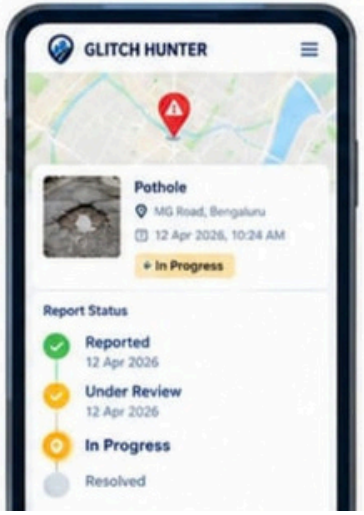
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SAMPLE UI DEVELOPMENT BY THE TEAM









REFERENCE APP & WEB

Smart Cities Mission
(Urban Development & Smart Cities)
Guidelines, best practices and resources for smart and sustainable cities.

MyGov Portal
(Citizen Participation)
Citizen engagement, feedback and grievance redressal platform.

Municipal Corporation Complaint Portals
Report civic issues, track status and get updates from your local municipality.

Civic Mobile Apps
Report, track and manage civic issues on the go.

REFERENCE ARTICLES & SOURCES

Ministry of Housing and Urban Affairs (MoHUA)
Policies, guidelines and initiatives for urban infrastructure and civic services.
mohua.gov.in

Smart Cities Mission
Information on smart city projects, standards and urban development.
smartcities.gov.in

OpenStreetMap
Open source map data for location and geospatial information.
openstreetmap.org

Municipal Grievance & Reporting Portals
City-specific portals for reporting civic issues and tracking progress.
e.g. mycity.gov.in (or your city portal)

GLITCH HUNTER web link: